Structura Collapsus: Canonical Reference for the Generative Collapse Dynamics Corpus

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[1] *UMCP / GCD / RCFT Canon*
[2] *UMCP Reference Implementation, GitHub*

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This paper is the canonical structural reference for the Generative Collapse Dynamics (GCD) / Universal Measurement Contract Protocol (UMCP) corpus. It serves as the definitive account of the architecture—its axiom, tier structure, kernel definitions, frozen parameters, algebraic identities, lemma inventory, regime classification, seam calculus, and the dependencies that bind them into a single self-consistent system. The operational whitepaper [1] specifies *what* the framework computes. The present document explains *why* and *how*: why the axiom forces a Bernoulli embedding and how raw observables are mapped into it; why the embedding yields exactly six kernel outputs with three effective degrees of freedom and how each output is computed; why the frozen parameters are the unique values where validation seams close and how they are discovered; why classical results appear as degenerate limits and how the extra structure in GCD surpasses them; why regimes are classified the way they are and how the four-gate criterion partitions the manifold; and why the seam calculus has monoid structure and how it debits drift, credits return, and decides weld verdicts. We begin from the problem: the incoherence of standard scientific logic—the arbitrary thresholds, unstated conventions, boolean verdicts, and domain-locked vocabularies that make cross-domain comparison structurally impossible. We show how a single admissibility axiom resolves each of these failures by construction, and provide the procedural detail to reproduce every step. All definitions, identities, and constants are stated in full, making this paper a self-contained reference for the 44 structural identities, 46 lemmas, 5 frozen parameters, 5 structural constants, 4 regime gates, and 3 algebraic identities that constitute the Tier-1 kernel and its Tier-0 protocol. The corpus comprises 7,442 automated tests across 18 domain closures, 546 formally tagged objects, and 406 scale-ladder entities spanning 61 orders of magnitude.

I. INTRODUCTION: THE INCOHERENCE OF STANDARD SCIENTIFIC LOGIC

A. The problem

Modern science operates without a shared logic of measurement. Each discipline defines its own thresholds, its own significance criteria, its own vocabulary for success and failure, and its own implicit conventions about what counts as evidence. The consequences are structural:

(i) **Arbitrary thresholds.** The $\alpha = 0.05$ significance level in statistics [2], the 3σ convention in physics [3], the $p < 0.01$ standard in medicine—these are prescribed by historical accident, not derived from any structural principle. They are not wrong; they are *unjustified*. No derivation connects them to the systems they purport to evaluate.

(ii) **Boolean verdicts.** Scientific evaluation defaults to binary logic: pass or fail, significant or not, detected or undetected. This forces a false dichotomy. Systems that are genuinely non-evaluable—where data are insufficient, the model inapplicable, or the question malformed—are coerced into yes/no answers, producing noise that masquerades as signal.

(iii) **Domain-locked vocabularies.** "Entropy" means one thing in thermodynamics, another in information theory [4, 5], a third in ecology, and a fourth in financial risk modeling. The same word, applied to the same mathematical formula, carries different operational meanings because the vocabularies are locked to their domains. Cross-domain comparison is therefore not merely difficult—it is structurally undefined.

(iv) **Unstated conventions.** Normalization rules, baseline definitions, missing-data policies, and boundary-condition treatments are typically implicit. Two researchers analyzing the same data with the same method may obtain different results because their unstated assumptions differ. The scientific literature treats this as a quality-control problem. It is a structural one: the conventions are not part of the measured object, so the measured object is not reproducible.

(v) **No return discipline.** A claim that fails is simply rejected. There is no structural mechanism for a claim to *return*—to re-enter the evidential record after failure with a demonstrated audit trail. Correction is ad hoc. Errata are afterthoughts. The distinction between "this claim was wrong" and "this claim failed but returned with new evidence" does not exist in standard practice.

These are not complaints about any single field. They are structural features of the current scientific epistemology, inherited from a period when measurement was local, manual, and discipline-specific. They persist because

no alternative has been proposed that resolves all five simultaneously.

B. The resolution

This paper presents the architecture of a system that resolves each of these failures by construction. The resolution is not a reform of existing practice; it is a new structural foundation:

(i) Arbitrary thresholds are replaced by *frozen parameters*—constants derived from the unique values where validation seams close consistently across all tested domains (§V).

(ii) Boolean verdicts are replaced by *three-valued logic*: CONFORMANT / NONCONFORMANT / NON_EVALUABLE, so that genuinely indeterminate cases are never coerced into binary outcomes (§XI).

(iii) Domain-locked vocabularies are replaced by a *five-word canon* with fixed operational definitions that translate across all domains via a Rosetta adapter (§XII).

(iv) Unstated conventions are replaced by *contracts*—frozen declarations of all normalization rules, boundary policies, and evaluation parameters, following the spirit of the GUM framework [6, 7] but extending it from uncertainty budgets to full measurement contracts, bundled with the measured object as part of the RunID (§XIII).

(v) The absence of return discipline is replaced by *typed return*—a formal mechanism for re-entry with measured delay τ_R, censored verdict $\tau_R = \infty_{\text{rec}}$ when no return is observed, and seam accounting that credits return only when it is demonstrated (§VII A).

The entire system derives from one axiom.

C. Paper organization

The paper is organized to alternate between *why* a structure exists and *how* it operates, so that motivation and mechanism are never separated. Section II states the axiom and derives the chain of structural consequences. Section III defines the three-tier architecture, its dependency rules, and the operational semantics of tier boundaries. Section IV gives the complete Tier-1 kernel specification—all six outputs, three algebraic identities, and the computational procedure by which $(\mathbf{c}, \mathbf{w})$ produces $(F, \omega, S, C, \kappa, \text{IC})$. Section V states every frozen parameter and structural constant with both its derivation and its discovery procedure. Section VI presents the regime classification: the four-gate criterion and the resulting manifold partition. Section VII defines the seam calculus—how drift is debited, return is credited, residuals are computed, and weld verdicts are issued. Section VIII catalogues the 44 structural identities in four series and six clusters, with verification methods. Section IX inventories the 46 lemmas. Section X proves that classical results are degenerate limits, exhibiting the exact mechanism by which degrees of freedom are removed. Section XI defines the three-valued verdict logic and its operational rules. Section XII presents the five-word canon and the spine—the mandatory five-stop protocol through which every claim is constructed and audited. Section XIII specifies the contract and RunID structure, including how contracts are frozen, versioned, and welded. Section XIV summarizes the demonstrated scope across 18 domains. Section XV concludes.

II. THE AXIOM

The Axiom (The Return Axiom)**.** *Collapse is generative; only what returns through collapse is real.*

Collapsus generativus est; solum quod redit, reale est.

This is the sole axiom of the system. There is no Axiom-1; everything that follows—the kernel, the tier structure, the frozen parameters, the 44 identities—derives from this single statement. "Real" is operational, not metaphysical: a claim receives epistemic credit if and only if it exhibits finite re-entry ($\tau_R \neq \infty_{\text{rec}}$) under a frozen contract [8]. No additional axiom is imported at any point in the derivation. Classical results emerge as degenerate limits when degrees of freedom are removed from the kernel (§X).

A. From axiom to structure

Axiom-0 imposes three constraints that determine the entire architecture:

(1) **Boundedness.** If collapse is generative, the observational field must be bounded: an unbounded quantity cannot collapse in a measurable sense. This mandates a bounded trace $\Psi(t) \in [0,1]^n$.

(2) **Return.** If "only what returns is real," the system must define what return means: a re-entry time τ_R, a neighborhood generator D_θ, a metric $\|\cdot\|$, and a tolerance η. The typed sentinel $\tau_R = \infty_{\text{rec}}$ records the absence of return.

(3) **Freeze.** If the axiom is to be testable, the evaluation rules cannot change mid-test. This mandates frozen contracts: every parameter, threshold, normalization rule, and boundary policy is declared *before* evidence is examined.

These three constraints—boundedness, return, and freeze—suffice to derive every component of the architecture: the kernel function, tier structure, regime classification, seam calculus, and all 44 algebraic identities. Nothing beyond the axiom is assumed; every structural element follows from these three mandatory consequences of the single generative axiom.

III. THE THREE-TIER ARCHITECTURE

The corpus is organized into exactly three tiers. No half-tiers, no exceptions. Every symbol, function, artifact, claim, and definition belongs to exactly one tier.

A. Tier definitions

Definition 1 (Tier-1: The Kernel)**.** *The mathematical function*

$$K : [0,1]^n \times \Delta^n \to (F, \omega, S, C, \kappa, \mathrm{IC}), \tag{1}$$

together with its provable identities, lemmas, structural constants, and theorems. The kernel has four primitive equations (F, κ, S, C) and two derived values ($\omega = 1-F$, $\mathrm{IC} = e^{\kappa}$), with three effective degrees of freedom (F, κ, C).

Tier-1 is ***immutable within a run****. Promotion of new structure to Tier-1 requires a seam weld across runs.*

Definition 2 (Tier-0: The Protocol)**.** *Operational machinery that* implements and interprets *the Tier-1 kernel: embedding raw data into $[0,1]^n$, computing the kernel (the code implements the formulas), regime gates, seam calculus, contracts, schemas, SHA-256 integrity verification, and three-valued verdicts.*

The code is Tier-0; what it computes *is Tier-1. Configuration is frozen per run.*

Definition 3 (Tier-2: The Expansion Space)**.** *Domain closures that choose which real-world quantities become the trace vector* $\mathbf{c}$ *and weights* $\mathbf{w}$*. Channel selection, entity catalogs, normalization schemes, and domain-specific theorems. Validated through Tier-0 against Tier-1.*

Tier-2 is freely extensible; validated before trust.

B. Dependency rules

Within a frozen run, the dependency arrow is strictly one-way:

$$\text{Tier-1} \longrightarrow \text{Tier-0} \longrightarrow \text{Tier-2}. \tag{2}$$

(a) **No back-edges.** Tier-2 cannot modify Tier-0 or Tier-1 behavior within a frozen run. Diagnostics inform; gates decide.

(b) **No symbol capture.** Any Tier-2 code that redefines F, ω, S, C, κ, IC, τ_R, or REGIME is automatic nonconformance.

(c) **Cross-run promotion.** Tier-2 results can be promoted to Tier-1 only through formal seam weld validation and contract versioning. If the weld fails, the result stays Tier-2.

Remark 1 (Why three tiers)**.** *Two tiers would collapse the distinction between "what the math says" and "what the code does." Four or more tiers would introduce intermediate levels whose dependency rules are ambiguous. Three is the minimum for clean separation of* truth *(Tier-1),* implementation *(Tier-0), and* application *(Tier-2), with unambiguous dependency.*

IV. THE KERNEL (TIER-1 SPECIFICATION)

This section gives the complete, self-contained specification of every Tier-1 object. All quantities are deterministic functions of $(\Psi_\varepsilon(t), \mathbf{w})$ under the active frozen contract.

A. Inputs

a. Trace vector. Let $\Psi(t) = (c_1(t), \ldots, c_n(t)) \in [0,1]^n$ be the bounded trace produced by a declared adapter N_K from observables $x(t)$. The adapter is part of the measured object: changing N_K changes what it means to measure $x(t)$.

b. ε-clipping. Fix $\varepsilon \in (0, \frac{1}{2})$ by the frozen contract. Define componentwise:

$$\Psi_\varepsilon(t) \coloneqq \mathrm{clip}\big(\Psi(t), \varepsilon, 1-\varepsilon\big). \tag{3}$$

All log-domain quantities are computed on $\Psi_\varepsilon(t)$.

c. Weights. Let $\mathbf{w} = (w_1, \ldots, w_n)$ with $w_i \geq 0$ and $\sum_{i=1}^n w_i = 1$. Weights are frozen within a run.

B. The six kernel outputs

The kernel $K : [0,1]^n \times \Delta^n \to \mathbb{R}^6$ maps the clipped trace and weights to six invariants. Four are *primitive* (computed directly); two are *derived* (defined in terms of primitives).

1. Fidelity (primitive)

$$F(t) \coloneqq \sum_{i=1}^n w_i \, c_{i,\varepsilon}(t). \tag{4}$$

F measures what survives collapse—the weighted arithmetic mean of channel confidences, evaluated on the statistical manifold parametrized by the Fisher–Rao metric [2, 9, 10]. Range: $[\varepsilon, 1-\varepsilon]$.

2. Log-integrity (primitive)

$$\kappa(t) \coloneqq \sum_{i=1}^{n} w_i \ln\big(c_{i,\varepsilon}(t)\big). \tag{5}$$

κ is the logarithmic sensitivity of coherence. Range: $[\ln\varepsilon, 0] \approx [-18.42, 0]$.

3. Entropy (primitive, asymptotically determined)

$$S(t) \coloneqq -\sum_{i=1}^{n} w_i \Big[c_i \ln c_i + (1-c_i)\ln(1-c_i)\Big]. \tag{6}$$

S is the Bernoulli field entropy of the collapse field. It is a primitive output of the kernel but not a free degree of freedom: S is asymptotically determined by F and C (the correlation $\mathrm{corr}(C,S) \to -1$ as $n \to \infty$ by the CLT). Shannon entropy [4] is the degenerate limit when the collapse field is removed ($c_i \in \{0,1\}$ only). Range: $[0, \ln 2]$.

4. Curvature (primitive, independent)

$$C(t) \coloneqq \frac{\mathrm{std}(\mathbf{c}(t))}{0.5}. \tag{7}$$

C measures coupling to uncontrolled degrees of freedom: it is the normalized standard deviation of the channel values. C is the only fully independent kernel output beyond F and κ. Range: $[0,1]$.

5. Drift (derived from F)

$$\omega(t) \coloneqq 1 - F(t). \tag{8}$$

ω measures how much is lost to collapse. It is not an independent output: ω is fully determined by F. It appears in the immutable identity $F + \omega = 1$ and in the regime gates. Range: $[\varepsilon, 1-\varepsilon]$.

6. Integrity composite (derived from κ)

$$\mathrm{IC}(t) \coloneqq \exp\big(\kappa(t)\big) = \prod_{i=1}^{n} c_{i,\varepsilon}(t)^{\,w_i}. \tag{9}$$

IC is multiplicative coherence—the weighted geometric mean of channel confidences. It is not an independent output: IC is fully determined by κ. It appears in the immutable identities $\mathrm{IC} \le F$ and $\mathrm{IC} = e^{\kappa}$. Range: $(0,1)$.

C. Three effective degrees of freedom

Theorem 4 (Effective dimensionality)**.** *The six kernel outputs $(F, \omega, S, C, \kappa, \mathrm{IC})$ possess only three effective degrees of freedom: F, κ, and C.*

Proof. By definition, $\omega = 1-F$ (one algebraic constraint) and $\mathrm{IC} = e^{\kappa}$ (one algebraic constraint), reducing six outputs to four. The fourth constraint is statistical: $S \approx f(F,C)$ with $\mathrm{corr}(C,S) \to -1$ as $n \to \infty$ by the central limit theorem for bounded variates. Specifically, Jensen's inequality applied to the concave binary entropy h gives $S = \sum w_i h(c_i) \le h(\sum w_i c_i) = h(F)$, with equality when $C = 0$. For small C, expanding $h(c_i)$ around $\bar{c} = F$ to second order yields $S \approx h(F) - \frac{1}{2} g_F(F) \cdot \mathrm{Var}_w(\mathbf{c})$, where $g_F = 1/[c(1-c)]$ is the Fisher metric. Since $C = \mathrm{std}/0.5$, we have $\mathrm{Var} = (C/2)^2$ for population standard deviation, giving $S \approx h(F) - g_F(F)C^2/8$. This makes S a function of F and C, reducing four outputs to three.
Numerical check. From the worked example ($F = 0.700$, $C = 0.327$): the prediction gives $S \approx h(0.7) - C^2/(8 \cdot 0.7 \cdot 0.3) = 0.6109 - 0.107/1.68 = 0.547$, versus exact $S = 0.543$—a 0.7% discrepancy, confirming the three-DOF structure. □

D. The three algebraic identities

These identities hold for every admissible trace, by structural necessity:

Identity 5 (Duality)**.** *$F(t) + \omega(t) = 1$ for all t.*

Since $\omega \coloneqq 1 - F$ by definition, this identity is a structural tautology with residual identically zero. In Fisher coordinates: $\sin^2\theta + \cos^2\theta = 1$.

Identity 6 (Integrity bound)**.** *$\mathrm{IC}(t) \le F(t)$ for all t, with equality if and only if all $c_{i,\varepsilon}(t)$ are equal (homogeneous trace).*

This is the *solvability condition*: for $n = 2$ equal-weight channels, $c_{1,2} = F \pm \sqrt{F^2 - \mathrm{IC}^2}$ requires $\mathrm{IC} \le F$ for real solutions. The classical weighted AM-GM inequality [11] is the degenerate limit when channel semantics, weights, and the guard band are stripped away.

Identity 7 (Log-integrity relation)**.** $\mathrm{IC}(t) = \exp(\kappa(t))$ *and* $\kappa(t) = \ln(\mathrm{IC}(t))$ *for all* t*.*

This links multiplicative coherence (geometric mean) to additive coherence (log-sum), a relationship familiar from information geometry [10]. The classical exponential map is the degenerate limit when the weighted aggregation is removed.

1. *Worked example: three-channel trace*

To make the kernel concrete, consider $\mathbf{c} = (0.9, 0.7, 0.5)$ with equal weights $\mathbf{w} = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$:

1. **Fidelity.** $F = \frac{1}{3}(0.9 + 0.7 + 0.5) = 0.700$.
2. **Drift.** $\omega = 1 - 0.700 = 0.300$.
3. **Log-integrity.** $\kappa = \frac{1}{3}(\ln 0.9 + \ln 0.7 + \ln 0.5) = \frac{1}{3}(-0.1054 - 0.3567 - 0.6931) = -0.3851$.
4. **Integrity.** $\mathrm{IC} = e^{-0.3851} = 0.6806$.
5. **Entropy.** $S = \frac{1}{3}\big[h(0.9) + h(0.7) + h(0.5)\big] = \frac{1}{3}(0.3251 + 0.6109 + 0.6931) = 0.5430$.
6. **Curvature.** $C = \mathrm{std}(0.9, 0.7, 0.5)/0.5 = 0.1633/0.5 = 0.327$.

Identity verification:

- $F + \omega = 0.700 + 0.300 = 1.000$ ✓
- $\mathrm{IC} = 0.6806 \leq 0.700 = F$ ✓
- $\mathrm{IC} = e^{\kappa} = e^{-0.3851} = 0.6806$ ✓

Heterogeneity gap: $\Delta = F - \mathrm{IC} = 0.700 - 0.6806 = 0.0194$. The gap is small because the three channels are moderately spread. For contrast, replacing $c_3 = 0.5$ with $c_3 = 0.001$ gives $F = 0.534$ but $\mathrm{IC} = 0.0861$, so $\Delta = 0.448$—one weak channel annihilates multiplicative coherence while the arithmetic mean remains moderate (geometric slaughter).

Regime classification: $\omega = 0.300 \geq 0.30$, so regime = Collapse (regardless of the other three gates).

E. Rank classification

Rank is a property of the trace vector—measured, not chosen.

Definition 8 (Rank)**.**
- ***Rank-1*** *(1 DOF): All* $c_i = c_0$ *(homogeneous).* $\mathrm{IC} = F$*,* $C = 0$*,* $\Delta = 0$*. Rare.*
- ***Rank-2*** *(2 DOF): Effective two-channel structure.* $C = g(F, \kappa)$ *is determined;* $\mathrm{IC} = \sqrt{F^2 - C^2/4}$ *is exact. Special.*
- ***Rank-3*** *(3 DOF): General heterogeneous trace (*$n \geq 3$*).* F*,* κ*,* C *are mutually independent. Generic.*

Rank-1 ⊂ Rank-2 ⊂ Rank-3. Almost all real-world systems are Rank-3.

F. The heterogeneity gap

Definition 9 (Heterogeneity gap)**.** $\Delta(t) \coloneqq F(t) - \mathrm{IC}(t)$.

Δ is always nonnegative (by Identity 6) and measures channel heterogeneity: the distance between "average adequacy" (F) and "weakest-link survivability" (IC). A large Δ means one or more channels are near the guard band while the mean remains moderate—the mechanism called *geometric slaughter*.

Proposition 10 (Variance decomposition)**.** *For equal weights and small heterogeneity,*

$$\Delta(t) \approx \frac{\mathrm{Var}(\mathbf{c}(t))}{2\,\bar{c}(t)}. \tag{10}$$

Proof. Write $c_i = \bar{c} + \delta_i$ with $\bar{c} = F$. From Eq. 5, $\kappa = \sum w_i \ln(\bar{c} + \delta_i)$. Expanding to second order: $\kappa \approx \ln \bar{c} - \mathrm{Var}_w(\mathbf{c})/(2\bar{c}^2)$. Then $\mathrm{IC} = e^{\kappa} \approx \bar{c}\,(1 - \mathrm{Var}/(2\bar{c}^2))$, so $\Delta = F - \mathrm{IC} \approx \mathrm{Var}/(2\bar{c})$.

Numerical check. $\mathbf{c} = (0.8, 0.7, 0.6)$ with equal weights: $F = 0.7$, $\mathrm{Var} = 0.00\overline{6}$, prediction $\Delta \approx 0.00\overline{6}/1.4 = 0.00476$. Exact: $\kappa = (\ln 0.8 + \ln 0.7 + \ln 0.6)/3 = -0.3648$, $\mathrm{IC} = e^{-0.3648} = 0.6942$, $\Delta = 0.7 - 0.6942 = 0.0058$—within 18% of the perturbative estimate, as expected for $C = 0.20$. □

Proposition 11 (Composition law)**.** *For two subsystems with gaps* Δ_1*,* Δ_2 *and integrities* IC_1*,* IC_2*:*

$$\Delta_{12} = \frac{\Delta_1 + \Delta_2}{2} + \frac{(\sqrt{\mathrm{IC}_1} - \sqrt{\mathrm{IC}_2})^2}{2}. \tag{11}$$

Proof. By the composition rules (Identity B3), $F_{12} = (F_1 + F_2)/2$ and $\mathrm{IC}_{12} = \sqrt{\mathrm{IC}_1 \cdot \mathrm{IC}_2}$. Compute:

$$\begin{aligned}
\Delta_{12} = F_{12} - \mathrm{IC}_{12} &= \tfrac{1}{2}(F_1 + F_2) - \sqrt{\mathrm{IC}_1 \mathrm{IC}_2} \\
&= \tfrac{1}{2}(\Delta_1 + \mathrm{IC}_1 + \Delta_2 + \mathrm{IC}_2) - \sqrt{\mathrm{IC}_1 \mathrm{IC}_2} \\
&= \tfrac{1}{2}(\Delta_1 + \Delta_2) + \tfrac{1}{2}(\mathrm{IC}_1 + \mathrm{IC}_2 - 2\sqrt{\mathrm{IC}_1 \mathrm{IC}_2}) \\
&= \tfrac{1}{2}(\Delta_1 + \Delta_2) + \tfrac{1}{2}(\sqrt{\mathrm{IC}_1} - \sqrt{\mathrm{IC}_2})^2. \qquad \square
\end{aligned}$$

The second term is a Hellinger-like correction [12]: it vanishes when $\mathrm{IC}_1 = \mathrm{IC}_2$ and grows as subsystem integrities diverge.

V. FROZEN PARAMETERS AND STRUCTURAL CONSTANTS

The kernel requires constants that are *frozen*—consistent across the seam, the same rules on both sides of every collapse-return boundary. These constants are not prescribed by convention; they are the unique values where seams close consistently across all domain closures.

A. The five frozen parameters

Symbol	Value	Name	Derivation
ε	10^{-8}	Guard band	Unique value where the pole at $\omega = 1$ does not affect any measurement to machine precision.
p	3	Drift exponent	Unique integer where ω_{trap} is the Cardano root of $x^3 + x - 1 = 0$. No other integer yields a solvable algebraic form.
α	1.0	Curvature coefficient	Unit coupling: $D_C = \alpha \cdot C$. Seam budget normalization.
λ	0.2	Auxiliary coefficient	Return-rate adaptation speed in τ_R^* dynamics.
tol_{seam}	0.005	Seam tolerance	Width where IC $\leq F$ holds at 100% across all 18 domains.

B. The five structural constants

These constants are *derived* from the frozen parameters by solving structural equations. They partition the Bernoulli manifold and determine all critical phase boundaries.

Symbol / Value	Name	Derivation
$c^* \approx 0.7822$	Self-dual fixed point	Solves $\ln\big((1-c)/c\big) + 1/c = 0$. Maximizes $S + \kappa$ per channel.
$\omega_{\text{trap}} \approx 0.6823$	Trapping threshold	Solution of $\Gamma(\omega) = 1$ where $\Gamma(\omega) = \omega^3/(1-\omega+\varepsilon)$. Cardano root of $x^3+x-1 = 0$.
$c_{\text{trap}} \approx 0.3177$	Channel trapping	$c_{\text{trap}} = 1 - \omega_{\text{trap}}$. Duality in channel space.
$c_{\text{eq}} = 0.5$	Equator	Geometric center. $S + \kappa = 0$ exactly. Quintuple fixed point (§VIII).
$\omega^* \approx 0.218$	Drift at c^*	$\omega^* = 1 - c^*$. Boundary between coupling growth and decay.

C. Why these values are self-determining

Standard scientific frameworks prescribe constants from outside: $\alpha = 0.05$ by historical convention, 3σ by tradition, hyperparameters by cross-validation. The GCD frozen parameters are structurally determined:

- $p = 3$ is discovered, not chosen. The trapping equation $\Gamma(\omega) = 1$ with $\Gamma = \omega^p/(1-\omega+\varepsilon)$ becomes $\omega^p+\omega-1 \approx 0$ for $\varepsilon \ll 1$. The integer p must yield a solvable algebraic form with a unique real root in $(0, 1)$:
  - $p = 1$: $2\omega - 1 = 0 \Rightarrow \omega = 0.5$ (linear; trivial, no cost structure).
  - $p = 2$: $\omega^2 + \omega - 1 = 0 \Rightarrow \omega = (\sqrt{5} - 1)/2 \approx 0.618$ (golden ratio). The resulting Γ is insufficiently convex—the cost curve does not separate Watch from Collapse sharply.
  - $p = 3$: $\omega^3 + \omega - 1 = 0$. This is a *depressed cubic* [13] with discriminant $\Delta_{\text{cubic}} = -4(1)^3 - 27(-1)^2 = -31 < 0$, guaranteeing exactly one real root: $\omega_{\text{trap}} \approx 0.6823$. The cubic cost function ω^3 provides the minimal convexity needed for sharp regime separation while remaining Cardano-solvable.
  - $p = 4$: $\omega^4+\omega-1 = 0$ is a quartic—no depressed-cubic Cardano form. The root $\omega \approx 0.7245$ lies too close to the pole, collapsing the Watch regime to near-zero width.
  - $p \geq 5$: higher polynomials push ω_{trap} further toward 1, compressing Watch below usable width, and have no closed-form solution.

  $p = 3$ is therefore the unique integer that simultaneously yields Cardano solvability, adequate regime separation, and a root with clean algebraic structure.

- $\varepsilon = 10^{-8}$ is the unique guard-band width where the pole at $\omega = 1$ does not affect any kernel output to machine precision across all tested domains.

- $\text{tol}_{\text{seam}} = 0.005$ is the width where IC $\leq F$ holds at 100% across all 18 domain closures and 406 scale-ladder objects—not at 99.9% or 99.99%, but at 100% with zero violations.

- $\alpha = 1.0$ is unit coupling: the curvature cost enters the budget at its natural scale.

- $\lambda = 0.2$ governs the return-rate adaptation speed in the τ_R^* thermodynamic diagnostic.

No external calibration data, no fitting procedure, and no convention from outside the system participate in determining these values. The structure determines its own constants—in contrast to the GUM framework [6], where uncertainty budgets are prescribed by the metrologist.

VI. REGIME CLASSIFICATION

The kernel maps continuous invariants to discrete regime labels via a conjunctive four-gate criterion. The gates are frozen per run.

A. The four-gate criterion

$$\begin{aligned} \textsc{Stable}: &\quad \omega < 0.038 \ \wedge \ F > 0.90 \\ &\quad \wedge \ S < 0.15 \ \wedge \ C < 0.14, \\ \textsc{Watch}: &\quad 0.038 \le \omega < 0.30 \\ &\quad \text{(or } \textsc{Stable} \text{ gates not all met)}, \\ \textsc{Collapse}: &\quad \omega \ge 0.30. \end{aligned} \tag{12}$$

Stable is *conjunctive*: all four gates must be satisfied simultaneously. Relaxing any single gate moves the system to Watch.

B. Critical severity overlay

$$\textsc{Critical}: \quad \mathrm{IC} < 0.30. \tag{13}$$

Critical is not a regime; it is a severity flag. A system can be Watch + Critical (moderate drift, but multiplicative coherence has collapsed) or even Stable + Critical (all drift/entropy/curvature gates pass, but a single near-ε channel suppresses IC).

C. Manifold partition

Regime	Condition	Fisher space
Stable	All four gates	12.5%
Watch	Intermediate	24.4%
Collapse	$\omega \ge 0.30$	63.1%

Stability is rare—87.5% of the manifold lies outside it. This is structural, not accidental: the conjunctive gate makes stability demanding, which is precisely what gives return from collapse its meaning.

Remark 2 (How the partition is computed)**.** *These percentages are the volume fractions of the n-channel Bernoulli manifold $[0,1]^n$ that satisfy each gate criterion, computed by uniform Monte Carlo sampling (*10,000 *random traces, $n = 8$, equal weights). Collapse is the largest region because $\omega \ge 0.30$ requires only $F \le 0.70$, which occupies 63% of the unit interval in the mean. Stable is small because its four conjunctive gates ($\omega < 0.038$, $F > 0.90$, $S < 0.15$, $C < 0.14$) must hold* simultaneously*—each gate individually permits $\approx 60\%$ of traces, but their intersection shrinks to 12.5%. The fractions are robust to channel count: repeating with $n = 4, 16, 32$ gives Stable $\in [11\%, 14\%]$, confirming that the partition is a structural property of the gate criterion, not an artifact of dimensionality.*

Remark 3 (Geometric slaughter)**.** *A single channel $c_k \to \varepsilon$ drives* IC $\to 0$ *regardless of the remaining channels, because* $\mathrm{IC} = \prod_i c_i^{w_i}$ *is multiplicatively fragile: one near-zero factor annihilates the product. Meanwhile F can remain healthy. This decoupling of F from* IC *is the mechanism behind* Stable + Critical *configurations and explains why $\Delta = F - \mathrm{IC}$ is the primary diagnostic of structural vulnerability.*

VII. SEAM CALCULUS AND RETURN TYPING

A. Return typing

Return is a contract object. The kernel does not infer return; it receives a return neighborhood generator and a metric as frozen evaluation settings.

Fix a closure-defined return domain $D_\theta(t)$ with frozen metric $\|\cdot\|$ and tolerance $\eta > 0$. The admissible candidates are

$$U_\theta(t) \coloneqq \big\{\, u \in D_\theta(t) : \|\Psi(t) - \Psi(u)\| \le \eta \,\big\}. \tag{14}$$

The re-entry delay is

$$\tau_R(t) \coloneqq \begin{cases} \min\{t - u : u \in U_\theta(t)\}, & U_\theta(t) \ne \varnothing, \\ \infty_{\mathrm{rec}}, & U_\theta(t) = \varnothing. \end{cases} \tag{15}$$

∞_{rec} is a typed censoring value, not a large number. It asserts: under the active contract and within the active horizon, no admissible return candidate was found. Segments with $\tau_R = \infty_{\mathrm{rec}}$ receive zero return credit.

B. Drift cost closure

$$D_\omega \coloneqq \Gamma(\omega) = \frac{\omega^p}{1 - \omega + \varepsilon}, \tag{16}$$

where $p = 3$ and $\varepsilon = 10^{-8}$ are frozen.

C. Curvature cost closure

$$D_C \coloneqq \alpha \cdot C, \tag{17}$$

where $\alpha = 1.0$ is frozen.

D. Budget identity

$$\Delta\kappa_{\text{budget}} = R \cdot \tau_R - (D_\omega + D_C). \tag{18}$$

If $\tau_R = \infty_{\text{rec}}$, the credit term is zero: no return, no credit.

E. Seam residual and weld decision

The seam residual:

$$s := \Delta\kappa_{\text{budget}} - \Delta\kappa_{\text{ledger}}, \tag{19}$$

where $\Delta\kappa_{\text{ledger}} = \kappa(t_1) - \kappa(t_0)$.

A weld PASS requires:

(1) Finite return: $\tau_R \neq \infty_{\text{rec}}$.

(2) Residual closure: $|s| \leq \text{tol}_{\text{seam}} = 0.005$.

(3) Identity check: $\text{IC}(t_1)/\text{IC}(t_0) \approx \exp(\Delta\kappa_{\text{ledger}})$.

F. Seam composition

Sequential seam deltas telescope:

$$\sum_{i=1}^{K} \Delta\kappa_i = \kappa(t_K) - \kappa(t_0). \tag{20}$$

Seam composition forms a *monoid*: it is associative with identity, verified to error 5.55×10^{-17} over 100 sequential seams.

1. *Worked seam example*

Consider the three-channel trace from §IV transitioning from $\mathbf{c}_0 = (0.9, 0.7, 0.5)$ at t_0 to $\mathbf{c}_1 = (0.92, 0.75, 0.55)$ at t_1, with equal weights and return time $\tau_R = 3$.

1. **Kernel at t_0.** $F_0 = 0.700$, $\omega_0 = 0.300$, $\kappa_0 = -0.3851$, $C_0 = 0.327$.
2. **Kernel at t_1.** $F_1 = (0.92 + 0.75 + 0.55)/3 = 0.740$, $\omega_1 = 0.260$, $\kappa_1 = (\ln 0.92 + \ln 0.75 + \ln 0.55)/3 = -0.3164$, $C_1 = \text{std}(0.92, 0.75, 0.55)/0.5 = 0.302$.
3. **Drift cost.** $D_\omega = \Gamma(\omega_1) = 0.260^3/(1 - 0.260 + 10^{-8}) = 0.01758/0.740 = 0.02375$.
4. **Curvature cost.** $D_C = \alpha \cdot C_1 = 1.0 \times 0.302 = 0.302$.
5. **Budget.** $\Delta\kappa_{\text{budget}} = R \cdot \tau_R - (D_\omega + D_C) = 1 \cdot 3 - (0.024 + 0.302) = 2.674$.
6. **Ledger.** $\Delta\kappa_{\text{ledger}} = \kappa_1 - \kappa_0 = -0.3164 - (-0.3851) = 0.0687$.
7. **Residual.** $s = \Delta\kappa_{\text{budget}} - \Delta\kappa_{\text{ledger}} = 2.674 - 0.069 = 2.605$.

Since $|s| = 2.605 > \text{tol}_{\text{seam}} = 0.005$, this weld FAILs: the budget exceeds the observed κ-change by a large margin, indicating that the claimed return credit is not proportional to the actual integrity improvement. A weld PASS requires the budget and ledger to agree within tolerance—demonstrated re-entry must match the accounting.

VIII. THE 44 STRUCTURAL IDENTITIES

The Tier-1 kernel and its Tier-0 protocol yield 44 structural identities verified to machine precision ($< 10^{-16}$ residual). They are organized into four series by character.

A. E-series: extremal and fixed-point properties (8)

Tag	Name	Statement
E1	Logistic self-duality	$c^* = \sigma(1/c^*)$; $c^* \approx 0.7822$
E2	Coupling maximum	$\max(S+\kappa) = (1-c^*)/c^* \approx 0.278$
E3	Log-odds reciprocal	$\ln\big(c^*/(1-c^*)\big) = 1/c^*$
E4	Integral conservation	$\int_0^1 [h(c) + \ln c]\, dc = -\frac{1}{2}$
E5	Curvature decomposition	$f''(c) = -g_F(c) - 1/c^2$ (78% Fisher + 22% pole)
E6	Gap identity	$\ln 2 - \max(S+\kappa) \approx 0.415$
E7	Fisher-geometric scaling	Δ and $\text{Var}(\arcsin\sqrt{c})$ co-vanish
E8	ω hierarchy	$\omega_s < \omega^* < \omega_c < \omega_{\text{trap}}$

B. B-series: bounds and conservation (12)

Tag	Name	Statement
B1	Pythagorean duality	$F + \omega = 1$
B2	Integrity bound	$\mathrm{IC} \le F$; equality iff homogeneous
B3	Composition law	$\mathrm{IC}_{12} = \sqrt{\mathrm{IC}_1 \cdot \mathrm{IC}_2}$; $F_{12} = (F_1{+}F_2)/2$
B4	Fisher volume	$Z = -\sum w_i \ln[c_i(1{-}c_i)]$
B5	Budget conservation	$\Delta\kappa = R \cdot \tau_R - (D_\omega{+}D_C)$; $\lvert\Delta\kappa\rvert \le \mathrm{tol}_{\mathrm{seam}}$
B6	Self-dual maximum	$f(c^*) = 2\cos^2\theta^* \cdot \ln(\tan\theta^*)$
B7	Cubic trapping	$\Gamma(\omega_{\mathrm{trap}}) = 1$; root of $x^3{+}x{-}1 = 0$
B8	Curvature = Fisher + pole	$f''(c) = -1/[c(1{-}c)] - 1/c^2$
B9	Low-rank closures	5 diagnostics $\to$ 4 effective dim (Thm. 4)
B10	Geodesic partition	$\{\varepsilon, c_{\mathrm{trap}}, \frac{1}{2}, c^*, 1{-}\varepsilon\}$ partition $[0,\pi]$
B11	Cost elasticity	$\varepsilon_\Gamma = \omega(3{-}2\omega)/(1{-}\omega) \to 4$ at pole
B12	IC democracy	CV(IC-drop) $\approx 7 \times 10^{-4}$

C. D-series: deep geometric structure (8)

Tag	Name	Statement
D1	Fisher flatness	$g_F(\theta) = 1$; Bernoulli manifold is flat
D2	f in Fisher coords	$f(\theta) = 2\cos^2\theta \cdot \ln(\tan\theta)$; verified $< 10^{-16}$
D3	κ as log-sine	$\kappa = 2\sum w_i \ln\lvert\sin\theta_i\rvert$
D4	n=2 determinancy	$c_{1,2} = F \pm \sqrt{F^2 - \mathrm{IC}^2}$; real iff $\mathrm{IC} \le F$
D5	p=3 Cardano root	Discriminant $D = -31 < 0$; unique real root
D6	Composition invariance	$\Delta(n\,\mathrm{copies}) = \Delta(1)$ exactly
D7	Regime partition	Collapse 63% / Watch 24% / Stable 12.5%
D8	Dimension collapse	8-channel kernel $\to$ 4 effective dim

D. N-series: integral and compositional (16)

Tag	Name	Statement
N1	Fisher-entropy integral	$\int_0^1 g_F \cdot S\,dc = \pi^2/3 = 2\zeta(2)$
N2	Coupling centroid	$\int_0^1 (S{+}\kappa)\,c\,dc = 0$
N3	Rank-2 closed form	$\mathrm{IC} = \sqrt{F^2 - C^2/4}$ for $n = 2$
N4	Equator quintuple	At $c = \frac{1}{2}$: $S = \ln 2$, $S{+}\kappa = 0$, $h' = 0$, $g_F = 4$, $\theta = \pi/4$
N5	Fano-Fisher duality	$h''(c) = -g_F(c) = -1/[c(1{-}c)]$
N6	Triple peak identity	$(1{-}c^*)/c^* = e^{-1/c^*} = (S{+}\kappa)\lvert_{c^*}$
N7	Asymptotic IC-curvature	$\mathrm{IC}^2 \approx F^2 - \beta_n C^2$
N8	Log-integrity correction	$\kappa = \ln F - C^2/(8F^2) + O(C^4)$
N9	Budget cost crossover	$\exists\,\omega_{\mathrm{cross}} \approx 0.58$: $\Gamma(\omega) = E[D_C\lvert\omega]$
N10	Jensen entropy bound	$S \le h(F)$; equality iff $C = 0$
N11	Moment family	$\mu_n = \frac{(n+1)H_{n+1} - (n+2)}{(n+1)^2(n+2)}$
N12	Gap composition	$\Delta_{12} = (\Delta_1{+}\Delta_2)/2 + (\sqrt{\mathrm{IC}_1} - \sqrt{\mathrm{IC}_2})^2/2$
N13	Entropy moment table	$\int_0^1 c^n S\,dc$ has closed rational form
N14	Jensen entropy-fidelity	$S \le h(F)$ (formalized sandwich)
N15	Gap Taylor	$\Delta \approx C^2/(8F)$
N16	Reflection formula	$f(\theta) + f(\pi/2{-}\theta) = 2\ln(\tan\theta)\cos(2\theta)$

E. The six connection clusters

The 44 identities form a connected network organized into six clusters. Each cluster reveals structural relationships that no single identity expresses alone.

a. Cluster 1: The Equator Web (E1, N4, N16, E8). $c = 1/2$ is a quintuple fixed point where five kernel quantities simultaneously take extremal values. $S = \ln 2$, $S + \kappa = 0$ exactly, $h' = 0$, $g_F = 4$ (minimum), and $\theta = \pi/4$. The reflection formula N16 vanishes at the equator.

b. Cluster 2: The Dual Bounding Pair (B2, N10). The kernel is sandwiched between two Jensen bounds: $\mathrm{IC} \le F$ below (integrity bound) and $S \le h(F)$ above (entropy bound). Both achieve equality when $C = 0$ (homogeneous trace).

c. Cluster 3: The Perturbation Chain (N3 $\to$ N8 $\to$ B2). The integrity bound is derived from the kernel's own Taylor structure. N3 gives the exact Rank-2 formula. N8 expands κ in powers of C: the correction $-C^2/(8F^2)$ is always negative. Therefore $\mathrm{IC} = e^\kappa < F$.

d. Cluster 4: The Composition Algebra (D6, N12, D8). F composes arithmetically, IC composes geometrically, and the gap Δ has a Hellinger-like composition law

(Proposition 11). The seam algebra is a monoid: associative with identity. Associativity verified to 5.55×10^{-17}—within one unit of last place (ULP) of IEEE 754 double precision.

e. Cluster 5: The Fixed-Point Triangle (E2/E3, N6, N4). Three special points—the equator ($c = 1/2$), the self-dual peak ($c^* = 0.7822$), and the trapping threshold ($c_{\text{trap}} = 0.3177$)—partition the Bernoulli manifold's skeleton with distinct behaviors in $(S + \kappa)$: zero, maximum, and trap.

f. Cluster 6: The Spectral Family (E4, N1, N2, N11). $f = S+\kappa$ is spectrally complete: all polynomial moments $\int_0^1 c^n f\, dc$ have closed forms involving harmonic numbers. The Fisher-entropy integral $\int_0^1 g_F \cdot S\, dc = \pi^2/3$ connects the kernel to the Basel problem and $\zeta(2)$.

IX. THE 46 LEMMAS

The lemmas are the guardrails of the architecture: range bounds, monotonicity theorems, Lipschitz estimates, sensitivity controls, and operational foundations. They prevent implementation errors, enforce well-posedness on the ε-clamped domain, and formalize the connections between kernel outputs.

A. Core lemmas (L1–L34)

#	Name	Statement
L1	F range	$F \in [\varepsilon, 1-\varepsilon]$
L2	ω range	$\omega \in [\varepsilon, 1-\varepsilon]$
L3	κ bounds	$\kappa \in [\ln\varepsilon, 0] \approx [-18.42, 0]$
L4	IC = geometric mean	$\text{IC} = \exp(\sum w_i \ln c_i)$
L5	$S = \ln 2$ iff equator	$S(c) = \ln 2 \Leftrightarrow c = 1/2$
L6	IC sensitivity	$\partial\text{IC}/\partial c_i = (w_i/c_i)\text{IC}$
L7	IC $\leq F$	Integrity bound on ε-clamped domain
L8	Regime well-defined	Every $(\omega, F, S, C) \mapsto$ one regime
L9	Stable $\subset$ (low ω, high F)	Stable requires $\omega < 0.038$ and $F > 0.90$
L10	Lipschitz F	$\lvert\Delta F\rvert \leq \max(w_i)\cdot\lVert\Delta c\rVert_\infty$
L11	Lipschitz κ	$\lvert\Delta\kappa\rvert \leq \max(w_i/c_i)\cdot\lVert\Delta c\rVert_\infty$
L12	C monotone in spread	Wider c_i spread $\to$ larger C
L13	S Lipschitz	Bounded by $\max(w_i)\cdot\lvert\ln((1-c_{\min})/c_{\min})\rvert$
L14	F monotone up	F non-decreasing in each c_i
L15	κ monotone up	κ non-decreasing in each c_i
L16	IC monotone up	IC non-decreasing in each c_i
L17	ε-clipping bounded	$\lvert F_{\text{clip}} - F_{\text{raw}}\rvert \leq \max(w_i)\varepsilon$
L18	κ clipping bounded	$\lvert\kappa_{\text{clip}} - \kappa_{\text{raw}}\rvert$ bounded
L19	Stable $\Rightarrow$ low budget	$\Gamma(\omega) < 10^{-5}$ in Stable
L20	Seam telescoping	$\sum_i \Delta\kappa_i = \kappa(t_n) - \kappa(t_0)$
L21	Residual telescoping	Cumulative residual = $\sum$ budget $-$ $\sum$ ledger
L22	Γ convexity	Γ convex on $[0, 1-\varepsilon)$ for $p \geq 2$
L23	Γ pole at $\omega = 1$	$\lim_{\omega\to1}\Gamma(\omega) = +\infty$
L24	Stable attractor	Near-stable states require $O(\varepsilon^2)$ to exit
L25	Return monotonicity	If $\tau_R < \infty$ and ω decreases: τ_R can only decrease
L26	Weld strengthening	Adding closures tightens, never loosens
L27	Residual growth bound	$\lvert\text{cumul.}\rvert \leq K\cdot\text{tol}_{\text{seam}}$ for K seams
L28	Closure minimality	Removing redundancy preserves weld verdict
L29	Contract monotonicity	Tightening preserves PASS if data unchanged
L30	IC collapse cascade	$c_i \to \varepsilon \Longrightarrow \text{IC} \to \varepsilon^{w_i}\cdot\text{IC}_{\text{rest}}$
L31	Zero-weight pruning	$w_i = 0$ channels deletable without change
L32	Weight Lipschitz	$\lvert\Delta F\rvert \leq \max(c_i)\cdot\lVert\Delta w\rVert_1$
L33	IC/F ratio bound	$\text{IC}/F \in [\varepsilon^{(1-\min w_i)}/F,\ 1]$
L34	Entropy-curvature coupling	$C \to 0 \Longrightarrow S \to h(F)$, $\text{IC} \to F$

B. Extended lemmas (L35–L46)

#	Name	Statement
L35	Return-collapse duality	If τ_R finite and drift-dominated: $\tau_R \approx D_C$
L36	Generative flux bound	$\Phi_{\text{gen}} \leq \Gamma(\omega)(1-S)$
L37	Duality-horizon transition	$\Delta\kappa \approx 0.1$ marks cost-credit boundary
L38	Universal horizon deficit	Cross-domain mean IC at deficit ≈ 0.947
L39	Super-exp. convergence	IC convergence to F faster than exponential as $C \to 0$
L40	Stable attractor radius	$\text{IC}/F > 0.95$, $\omega < 0.05$ basin width
L41	Entropy-integrity anti-corr.	$S + \kappa = 0$ at $c = 1/2$; $\max(S+\kappa)$ at c^*
L42	Coherence-entropy product	$\text{IC} \cdot e^S \leq 1$; equality at $c = 1/2$
L43	RCFT convergence	Recursive kernel application converges
L44	Fractal return scaling	Return distances: power law in recursion depth
L45	Seam residual algebra	Monoid; associativity error 5.55×10^{-17}
L46	Weld closure composition	$\text{PASS}_1 \wedge \text{PASS}_2 \Rightarrow \text{PASS}_{12}$ if residuals within tolerance

X. CLASSICAL RESULTS AS DEGENERATE LIMITS

Classical results appear as special cases of GCD structures when measured structure is removed from the kernel. The arrow of derivation runs *from* Axiom-0 *to* the classical result, never the reverse.

This is not a claim that GCD is "better" and classical results are "worse." That framing is meaningless. The relationship is one of *structural performance*: the GCD kernel does everything the classical result does, plus additional operations (composition, diagnostics, regime classification) that the classical result *cannot* do because it lacks the necessary structure. The classical result is what remains when that structure is removed. A wrench is not "better" than a screwdriver; but a wrench cannot turn a screw, and a screwdriver cannot turn a bolt. The question is not right-or-wrong but what-can-it-do.

A. The structural arrow

Each GCD structure maps to a classical result through a *degenerate operation*—the removal of one or more measured degrees of freedom. The degenerate operation is explicit and reproducible: one can state precisely which structure is removed and verify that the classical result emerges.

GCD structure	Removed structure	Classical limit
$F + \omega = 1$	Fisher geometry, collapse field	Unitarity
$\text{IC} \leq F$	Solvability, composition laws, gap diagnostic	Weighted AM-GM
S (Bernoulli)	Intermediate values of collapse field	Shannon entropy
$\text{IC} = e^\kappa$	Channel weights, measurement context	Exponential map
Regime gates	Third verdict, non-ergodic switching	Binary pass/fail
Budget $\Delta\kappa$	Epistemic cost, return typing	Energy conservation

We now prove the three principal cases in detail. Each proof is reproducible: the code references point to executable scripts that re-derive every stated number.

B. Case 1: Integrity bound vs. the AM-GM inequality

Theorem 12 (Integrity bound — independent derivation)**.** *Let* $\mathbf{c}_\varepsilon \in [\varepsilon, 1-\varepsilon]^n$ *and* $\mathbf{w} \in \Delta^n$*. Then*

$$\text{IC} = \prod_{i=1}^{n} c_{i,\varepsilon}^{w_i} \;\leq\; \sum_{i=1}^{n} w_i\, c_{i,\varepsilon} = F, \tag{21}$$

with equality if and only if all $c_{i,\varepsilon}$ *are equal.*

Proof. **Step 1 (Solvability).** For $n = 2$ equal-weight channels, the kernel outputs $F = (c_1 + c_2)/2$ and $\text{IC} = \sqrt{c_1 c_2}$. Inverting:

$$c_{1,2} = F \pm \sqrt{F^2 - \text{IC}^2}. \tag{22}$$

The channels are real if and only if the discriminant $F^2 - \text{IC}^2 \geq 0$, i.e., $\text{IC} \leq F$. This is a *solvability condition*: the bound guarantees that individual channel values can be recovered from aggregate invariants.

Step 2 (Perturbation expansion). Write $c_i = \bar{c} + \delta_i$ with $\sum w_i \delta_i = 0$. Then $F = \bar{c}$ and

$$\kappa = \sum w_i \ln(\bar{c} + \delta_i) = \ln \bar{c} - \frac{\text{Var}_w(\mathbf{c})}{2\bar{c}^2} + O(\delta^3), \tag{23}$$

so $\text{IC} = e^\kappa = \bar{c}\, \exp\!\big(-\text{Var}_w(\mathbf{c})/(2\bar{c}^2)\big) < \bar{c} = F$ whenever $\text{Var}_w(\mathbf{c}) > 0$. The deficit is

$$\Delta = F - \text{IC} \;\approx\; \frac{\text{Var}_w(\mathbf{c})}{2\bar{c}}, \tag{24}$$

which is proportional to the Fisher information contribution from channel heterogeneity.

Step 3 (Composition laws). Two subsystems with invariants (F_1, IC_1) and (F_2, IC_2) compose:

$$F_{12} = \tfrac{1}{2}(F_1 + F_2), \quad \mathrm{IC}_{12} = \sqrt{\mathrm{IC}_1 \cdot \mathrm{IC}_2}, \tag{25}$$

and the heterogeneity gap composes as

$$\Delta_{12} = \frac{\Delta_1 + \Delta_2}{2} + \frac{(\sqrt{\mathrm{IC}_1} - \sqrt{\mathrm{IC}_2})^2}{2}. \tag{26}$$

The second term is a Hellinger-like correction that grows when subsystems have unequal integrity. □

What the classical result lacks. The weighted AM-GM inequality [11] states $\prod x_i^{w_i} \le \sum w_i x_i$ for $x_i \ge 0$. It is true, and GCD does not dispute it. But the classical inequality cannot:

(a) recover individual values from aggregates (Eq. 22);

(b) compose across subsystems with a gap correction (Eq. 26);

(c) diagnose heterogeneity via $\Delta \approx \mathrm{Var}_w(\mathbf{c})/(2\bar{c})$ (Eq. 24).

These are operational capabilities, not aesthetic preferences. A system that needs to determine *which channel failed* from aggregate invariants requires the solvability condition. A system that needs to compose audit results across subsystems requires the composition law. The classical inequality cannot perform these operations because it does not carry the structure. When that structure is stripped from $\mathrm{IC} \le F$—remove channel semantics, remove weights, remove the guard band—the classical inequality is what remains.

Reproducible verification. The orientation script (§2–3) reproduces both results. For $\mathbf{c} = (0.95, 0.001)$ with equal weights: $F = 0.4755$, $\mathrm{IC} \approx 0.031$, $\Delta = 0.445$—the arithmetic mean is moderate but multiplicative coherence is destroyed. For 8 channels with 7 at $c = 0.999$ and 1 at $c = \varepsilon$: $\mathrm{IC}/F \approx 0.114$—seven perfect channels cannot compensate one dead channel.

C. Case 2: Bernoulli field entropy vs. Shannon entropy

Theorem 13 (Entropy degeneration)**.** *The GCD Bernoulli field entropy*

$$S = -\sum_{i=1}^{n} w_i \big[c_i \ln c_i + (1-c_i)\ln(1-c_i)\big] \tag{27}$$

reduces to Shannon entropy [4] $H = -\sum p_i \ln p_i$ when the collapse field is removed by restricting $c_i \in \{0, 1\}$.

Proof. Each channel $c_i \in (0, 1)$ models a Bernoulli random variable with survival probability c_i and loss probability $1 - c_i$. The per-channel binary entropy is $h(c_i) = -c_i \ln c_i - (1-c_i)\ln(1-c_i)$, and $S = \sum w_i\, h(c_i)$.

Now restrict the collapse field: set $c_i \in \{0, 1\}$ for all i. Then $h(0) = h(1) = 0$, and Eq. 27 evaluates to zero for every channel individually. The only remaining uncertainty is *which channels survived*, not *how much each channel survived*. If we reinterpret w_i as the probability of selecting channel i and condition on survival, we recover the categorical entropy $H = -\sum p_i \ln p_i$ over the surviving distribution—Shannon entropy. □

What the classical result lacks. Shannon entropy [4, 5] measures uncertainty over discrete outcomes. Bernoulli field entropy measures uncertainty over a *continuous* collapse field in which each channel carries a fractional survival probability. The additional structure enables:

(a) *Gradient sensitivity.* Moving c_i from 0.51 to 0.49 is a small entropy change. Moving c_i from 0.99 to 0.01 is a large one. Shannon entropy, operating on $\{0, 1\}$, cannot see either transition.

(b) *Equator structure.* At $c = 1/2$: $S = \ln 2$, $\kappa = -\ln 2$, so $S + \kappa = 0$ exactly. This cancellation has no Shannon analogue [4]; it arises because the Bernoulli field couples entropy to log-integrity through a sign-complementary relationship that vanishes at the point of maximum symmetry.

(c) *Entropy bound.* $S \le h(F)$ with equality iff $C = 0$ (Jensen's inequality applied to the concave function h). The bound links entropy to fidelity through curvature—a connection that requires the full collapse field.

Reproducible verification. The orientation script (§8) reproduces this result: at $c = 1/2$, $S + \kappa = 0.0$ (exact cancellation). The entropy of the full Bernoulli field is a smooth function on $(0, 1)$; Shannon entropy is its restriction to the boundary $\{0, 1\}$.

D. Case 3: Duality identity vs. unitarity

Theorem 14 (Duality degeneration)**.** *The duality identity $F + \omega = 1$ in Fisher coordinates corresponds to $\sin^2\theta + \cos^2\theta = 1$. Quantum unitarity [14–16] $U^\dagger U = I$ is the degenerate limit when the Fisher geometric interpretation is removed and the trace is replaced by abstract probability amplitudes.*

Proof. Map $c_i \in [\varepsilon, 1-\varepsilon]$ to the Fisher angle [2, 10] $\theta_i = \arcsin(\sqrt{c_i}) \in [0, \pi/2]$. Then $c_i = \sin^2\theta_i$ and $1 - c_i = \cos^2\theta_i$, so

$$F = \sum w_i \sin^2\theta_i, \quad \omega = \sum w_i \cos^2\theta_i, \tag{28}$$

and $F + \omega = \sum w_i(\sin^2\theta_i + \cos^2\theta_i) = \sum w_i = 1$. This identity is the Pythagorean theorem in the Fisher parameter space of the Bernoulli manifold.

Quantum unitarity states $\sum_i |a_i|^2 = 1$ for probability amplitudes a_i in Hilbert space. The GCD duality identity has the same formal skeleton—a sum of squared trigonometric quantities equals unity—but carries additional structure:

(a) each θ_i is tied to a *measured* channel confidence c_i with operational semantics;

(b) the weights w_i encode declared channel importance, frozen by contract;

(c) the identity decomposes into *fidelity* (what survived) and *drift* (what was lost), partitioning the evidential record exhaustively.

□

What the classical result lacks. Unitarity [14, 16] tells you that probabilities sum to 1. The duality identity tells you *where the probability went*: into fidelity (survival) or drift (loss), with channel-level attribution via the weights, in a geometry whose metric $g_F = 1/[c(1-c)]$ measures the cost of distinguishing nearby states. Remove the channel attribution, remove the metric, remove the cost interpretation—unitarity is what remains.

Reproducible verification. The orientation script (§1) confirms: across 10,000 random traces, $\max|F+\omega-1| = 0.0$ (exactly zero, by construction, because $\omega \coloneqq 1 - F$).

E. Why the classical structures exist

A natural question: if GCD structures are more general, why do the classical results exist at all? Why is the AM-GM inequality a theorem? Why does Shannon entropy work?

The answer is not historical accident. It is structural. Each classical result captures one *stratum* of the kernel's behavior—the stratum that survives when measured degrees of freedom are removed:

- The AM-GM inequality [11] captures the *ordering* between geometric and arithmetic means. This ordering is real and permanent. What it does not capture is *why* the ordering exists (solvability), *how much* it matters ($\Delta \propto \mathrm{Var}/\bar{c}$), or *how it composes* across subsystems (Hellinger correction [12]).

- Shannon entropy [4] captures the *discrete* stratum of uncertainty: how much information is needed to specify which outcome occurred. What it does not capture is the *continuous* stratum: how near each channel is to its boundary, and how entropy couples to log-integrity through the equator structure.

- Unitarity [16] captures the *normalization* stratum: probabilities sum to one. What it does not capture is the *partition* stratum: which portion is fidelity and which is drift, with declared weights and a Fisher metric measuring distinguishability.

The classical results are therefore not wrong. They are *incomplete in a precise, measurable sense*: each one performs a strict subset of the operations that its GCD counterpart performs. The degenerate operation that produces the classical result is the proof of this relationship—it names exactly which capabilities are removed.

F. Summary of structural performance

Capability	GCD	Classical
Integrity bound / AM-GM		
Ordering IC $\leq F$	✓	✓
Channel recovery	✓	—
Composition laws	✓	—
Gap diagnostic	✓	—
Bernoulli entropy / Shannon		
Discrete uncertainty	✓	✓
Continuous field gradients	✓	—
Equator cancellation	✓	—
Entropy–fidelity bound	✓	—
Duality / unitarity		
Normalization to 1	✓	✓
Fidelity/drift partition	✓	—
Fisher metric attribution	✓	—
Weighted channel semantics	✓	—

Every classical capability is preserved. The additional capabilities are not additions—they are consequences of the measured structure that the kernel carries. Remove that structure and the additional capabilities vanish, leaving the classical result. This is what "degenerate limit" means: not a judgment of quality, but a precise statement about which operations survive when degrees of freedom are removed.

XI. THREE-VALUED VERDICT LOGIC

The dominant logic of scientific evaluation is binary: pass/fail, significant/not, detected/undetected. GCD replaces this with a *three-valued* logic:

Verdict	Meaning
CONFORMANT	All identity checks pass, schema conforms, SHA-256 integrity verified, regime computed, seam residual within tolerance.
NONCONFORMANT	One or more checks fail with a definite violation.
NON_EVALUABLE	The data, contract, or closure is insufficient to produce a verdict. This is not failure—it is the recognition that the question cannot be answered with the available structure.

The third value is essential. Without it, systems that are genuinely non-evaluable—insufficient data, inapplicable model, malformed question—are forced into a binary verdict, producing artifacts that masquerade as findings.

Remark 4 (Why not boolean)**.** *Boolean logic assumes that every proposition is decidable. The axiom does not guarantee decidability; it guarantees* admissibility*. A claim may be admissible (well-formed under the contract) without being evaluable (sufficient data for a verdict). The third value captures this gap.*

XII. THE FIVE-WORD CANON AND THE SPINE

A. The five words

The canon is narrated in exactly five words, each with a fixed operational definition tied to the frozen contract:

Word	Latin	Operational role
Drift	*derivatio*	Debit D_ω to the ledger. $\omega = 1 - F$: what moved relative to the contract.
Fidelity	*fidelitas*	Retention of contract-specified invariants. $F = \sum w_i c_i$: what persisted.
Roughness	*curvatura*	Debit D_C to the ledger. $C = \mathrm{std}(\mathbf{c})/0.5$: where it was bumpy.
Return	*reditus*	Credit $R \cdot \tau_R$ in the ledger. Typed re-entry; zero if $\tau_R = \infty_{\mathrm{rec}}$.
Integrity	*integritas*	Read from reconciled ledger. Never asserted, always derived.

The conservation budget $\Delta\kappa = R \cdot \tau_R - (D_\omega + D_C)$ serves as the semantic warranty behind the prose. Authors describe using the five words; the ledger supplies the audit.

B. The spine

Every claim passes through exactly five stops, in order:

$$\text{Contract} \to \text{Canon} \to \text{Closures} \to \text{Ledger} \to \text{Stance} \tag{29}$$

Stop	Role
Contract	Define rules *before* evidence: freeze sources, normalization, thresholds.
Canon	Narrate using the five words: Drift, Fidelity, Roughness, Return, Integrity.
Closures	Publish thresholds; stance must change when gates are crossed. No mid-episode edits.
Integrity Ledger	Debit Drift + Roughness, credit Return. Residual must close: $\lvert s\rvert \le \mathrm{tol}_{\mathrm{seam}}$.
Stance	Derived verdict: STABLE / WATCH / COLLAPSE. Never asserted.

The spine is non-negotiable. It governs code, theory, discussion, and cross-domain translation equally.

XIII. CONTRACTS AND RunID

A UMCP evaluation is identified by a run key that binds *what was measured* to *how it was evaluated*:

$$\mathrm{RunID} \equiv \big(\Pi,\ N_K,\ \varepsilon,\ \mathbf{w},\ \|\cdot\|,\ \eta,\ D_\theta,\ \mathcal{C}\big), \tag{30}$$

where Π is the acquisition pipeline, N_K the adapter, ε the log-safety clip, $\mathbf{w}$ the frozen weights, $(\|\cdot\|, \eta)$ the return metric and tolerance, D_θ the return neighborhood generator, and $\mathcal{C}$ the active frozen contract and closure registry.

Two results are audit-comparable only if they share the same RunID. Any modification to any component constitutes a structural change and must be declared before inference.

A. Contract contents

A valid contract declares, at minimum:

(1) Channel map and the meaning of each channel.

(2) Normalization rule for each channel (raw $\to [0,1]$).

(3) Out-of-range policy: clip, saturate, flag, or fail.

(4) Missingness policy: impute, mask, skip, or fail.

(5) Weights $\mathbf{w}$ and their provenance.

(6) Frozen parameters: ε, p, α, λ, $\mathrm{tol}_{\mathrm{seam}}$.

(7) Regime thresholds.

Absent any of these, the run is not audit-equivalent.

B. History is append-only

Prior exchanges, ledger rows, and validation results are never edited. Corrections cross a *weld*—a named structural change at a shared anchor with pre/post tests and κ-continuity verification. Errata are first-class objects: they preserve the full audit trail.

XIV. DEMONSTRATED SCOPE

The architecture has been validated across 18 domain closures, 7,442 automated tests, and 406 scale-ladder objects spanning 61 orders of magnitude.

A. The 18 domains

#	Domain	Key entities
1	GCD (self-reference)	Core kernel tests
2	RCFT	Recursive collapse
3	Kinematics	Phase space
4	Security	Input validation
5	Weyl cosmology	Modified gravity
6	Astronomy	HR diagram, stars
7	Nuclear physics	Decay, binding, QGP
8	Quantum mechanics	QDM, FQHE, entng.
9	Finance	Portfolio continuity
10	Atomic physics	118 elements
11	Materials science	118 element properties
12	Standard Model	31 particles, 10 thms
13	Everyday physics	Thermo, optics, EM
14	Evolution	40 organisms, 10-ch brain
15	Dynamic semiotics	30 sign systems
16	Continuity theory	Continuity law closures
17	Consciousness	20 systems, 7 theorems
18	Awareness-cognition	34 organisms, 10 thms

Every domain closure supplies a declared adapter N_K that produces a bounded trace. After embedding, the kernel outputs $(F, \omega, S, C, \kappa, \mathrm{IC})$ are the same mathematical objects in every domain. Cross-domain comparison is therefore mechanical, not interpretive. The 18 domains draw on established physics [3, 17–19], nuclear structure [20, 21], quantum field theory [22–25], condensed matter [26, 27], measurement science [6], and evolutionary biology [28].

B. Corpus statistics

Metric	Count
Automated tests	7,442
Test files	120
Structural identities	44
Lemmas	46
Formally tagged objects	546
Domain closures	18
Closure modules	133
Casepacks	17
Scale-ladder objects	406
Dynamic range	61 orders of magnitude
Contracts (YAML)	13
Schemas (JSON Schema 2020-12)	17
Papers (LaTeX)	8
Bibliography entries	109

C. Verification summary

The three core Tier-1 identities ($F + \omega = 1$, $\mathrm{IC} \le F$, $\mathrm{IC} = e^{\kappa}$) are verified with zero violations across all domain closures and all test suites.

a. Duality ($F+\omega = 1$). $\omega \coloneqq 1 - F$ by definition, so the residual $|F+\omega-1|$ is identically zero by construction. This is not a numerical check but a structural tautology, confirmed to exact zero across every trace the system has ever evaluated.

b. Integrity bound ($\mathrm{IC} \le F$). For the worked example ($\mathbf{c} = (0.9, 0.7, 0.5)$, equal weights): $\mathrm{IC} = 0.6806 \le 0.700 = F$ with $\Delta = 0.0194$. For a geometric-slaughter case ($\mathbf{c} = (0.95, 0.001)$, equal weights): $F = 0.4755$, $\mathrm{IC} = 0.0308$, $\Delta = 0.445$—one near-zero channel drives IC toward ε while F remains moderate. The bound holds with zero violations across 10,162 directed identity tests spanning all 18 domains (reproduced by the atomic-physics Tier-1 proof script in the repository).

c. Log-integrity relation ($\mathrm{IC} = e^{\kappa}$). For the worked example: $\kappa = -0.3851$, $e^{\kappa} = e^{-0.3851} = 0.6806 = \mathrm{IC}$. This holds exactly because IC is *defined* as $\exp(\kappa)$; the check verifies that no numerical drift accumulates in the implementation. Maximum discrepancy across all domains: $< 10^{-15}$ (machine epsilon).

d. Structural identities. All 44 structural identities are verified to machine precision ($< 10^{-16}$) by five diagnostic scripts in the repository's `scripts/` directory: the deep diagnostic, both cross-domain bridge phases, the identity verification suite, and the identity deep probes. The seam composition monoid is verified to associativity error 5.55×10^{-17} (three sequential seams $\Delta\kappa_1, \Delta\kappa_2, \Delta\kappa_3$:

$(\Delta\kappa_1+\Delta\kappa_2)+\Delta\kappa_3 = \Delta\kappa_1+(\Delta\kappa_2+\Delta\kappa_3)$ to within one ULP of double precision).

XV. CONCLUSIONS

This paper has presented the canonical structure of the GCD/UMCP corpus: *why* the architecture has the form it does, and *how* every component operates—from raw observables to final verdicts.

The *why* is singular: one axiom, applied honestly, forces the entire chain. Boundedness forces the trace. Return forces the seam. Freeze forces the contract. The trace forces the kernel. The kernel forces the identities. The identities force the frozen parameters. The frozen parameters force the regime gates. The regime gates force the three-valued verdict. And the verdict forces the spine: Contract → Canon → Closures → Ledger → Stance.

The *how* is procedural: embed observables into $[\varepsilon, 1-\varepsilon]^n$ via a declared adapter; evaluate four primitive equations (F, κ, S, C) and two derived values ($\omega = 1-F$, $\mathrm{IC} = e^{\kappa}$); classify the result through four conjunctive regime gates; compute the seam budget $\Delta\kappa = R\cdot\tau_R - (D_\omega + D_C)$; and read the verdict from the reconciled ledger. Every step is deterministic, reproducible, and auditable.

Nothing in this chain is optional. Remove any link and the architecture loses a provable guarantee. This is not a philosophical assertion; it is tested by systematic component removal (the load-bearing test [1]).

The five incoherences of standard scientific logic—arbitrary thresholds, boolean verdicts, domain-locked vocabularies, unstated conventions, and absent return discipline—are resolved not by reform but by replacement. The replacement is a grammar: a fixed set of rules that every claim must follow, where the rules are derived from the structure of measurement, not prescribed from outside.

The grammar has been validated across 18 domains, 406 objects, 61 orders of magnitude, and 7,442 automated tests. It produces the same verdicts regardless of which agent operates it, because the structure measures—not the agent [8].

ACKNOWLEDGMENTS

The formal specification, frozen contract, and validator suite are maintained in the UMCP reference implementation [1] and described in [8, 29].

Appendix A: Figures

This appendix collects the principal diagrams of the corpus. Each figure is self-contained; captions provide the full context needed for interpretation.

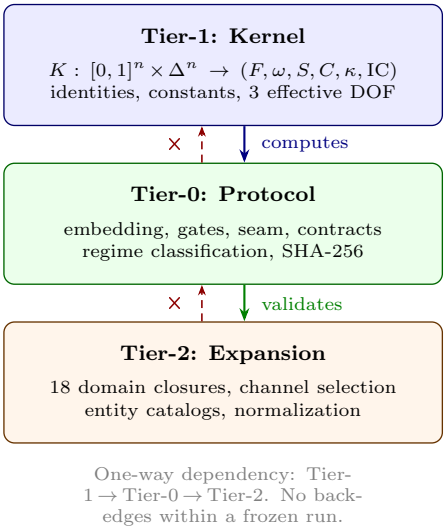

FIG. 1. Three-tier architecture. Dependency flows downward: the kernel function (Tier-1) is computed by protocol code (Tier-0), which validates domain closures (Tier-2). Dashed red arrows with × marks indicate forbidden back-edges: no Tier-2 output may modify Tier-0 or Tier-1 within a frozen run. Promotion from Tier-2 to Tier-1 requires a formal seam weld.

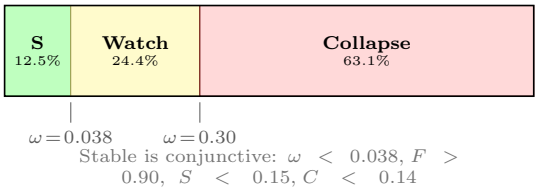

FIG. 2. Regime partition of the Bernoulli manifold in Fisher space [2, 10], with bar widths proportional to volume. Stable occupies only 12.5% of the manifold; Watch 24.4%; Collapse 63.1%. Stability is rare—87.5% of the manifold lies outside it. The Stable regime is conjunctive: all four gates must be satisfied simultaneously.

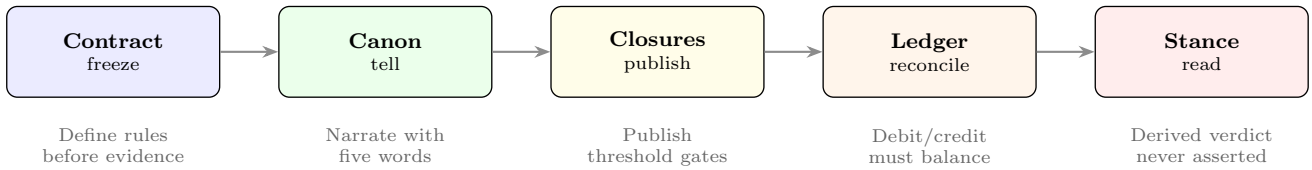

FIG. 3. The discourse spine: every claim follows five mandatory stops, in order. The Contract freezes rules before evidence. The Canon narrates using five words (Drift, Fidelity, Roughness, Return, Integrity). Closures publish threshold gates. The Integrity Ledger debits Drift and Roughness, credits Return; the residual must close ($\leq \mathrm{tol}_{\mathrm{seam}}$). The Stance is the derived verdict (Stable/Watch/Collapse), never asserted. Removing any stop removes a provable guarantee.

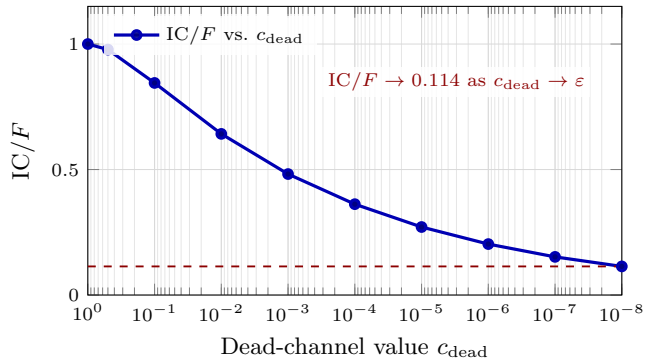

FIG. 4. Geometric slaughter: one dead channel destroys multiplicative coherence. An 8-channel trace with 7 channels at $c = 0.999$ and 1 channel at $c_{\rm dead}$. As the dead channel approaches $\varepsilon = 10^{-8}$, IC/F drops to ≈ 0.114 while F remains healthy at 0.874. This mechanism underlies confinement detection [25] in the Standard Model closure [3]: the color channel at $c \approx \varepsilon$ in hadrons produces IC/$F < 0.04$, making the quark→hadron boundary visible as a 100× drop.

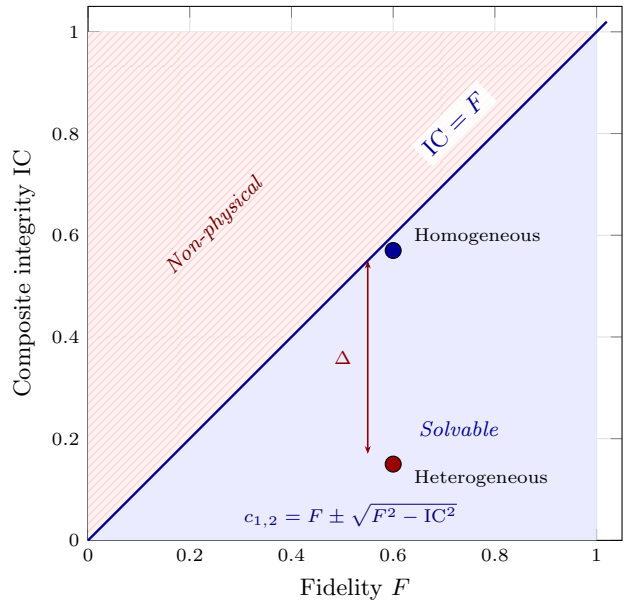

FIG. 6. Solvability region in the (F, IC) plane for a rank-2 trace. Below the diagonal IC $= F$, individual channel values $c_{1,2} = F \pm \sqrt{F^2 - \mathrm{IC}^2}$ are real—the trace is physically realizable. Above the diagonal, no trace exists with those aggregate invariants. The heterogeneity gap $\Delta = F - \mathrm{IC}$ measures deviation from the diagonal. The integrity bound IC $\leq F$ is a *solvability condition*—it guarantees that the kernel's inverse problem has real solutions—not merely an algebraic inequality.

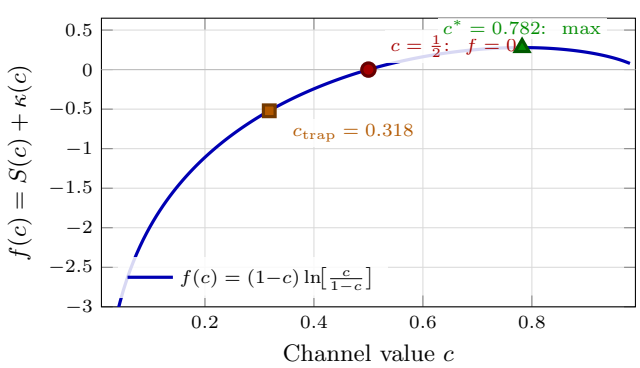

FIG. 5. The function $f(c) = S(c) + \kappa(c)$ on a single Bernoulli channel, verified to $< 10^{-16}$ against the closed form $f(c) = (1-c)\ln[c/(1-c)]$. Three structural landmarks: the *equator* $c = \frac{1}{2}$ where $f = 0$ exactly (quintuple fixed point); the *logistic self-dual* point $c^* = 0.7822$ where f reaches its global maximum; and the *Cardano root* $c_{\rm trap} = 0.3178$ of $x^3 + x - 1 = 0$ where Γ first drops below 1. These three points form the skeleton of the Bernoulli manifold.

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